

Exercices for the candidate as Backend Developer

Exercise 1: URL Shortener

Implement a Python command line tool that performs URL shortening and expansion.

The tool has two primary functions:

1. Given a complete URL, return a shortened URL.
2. Given a shortened URL, return the complete URL.

The shortened URLs has an expiration time of N seconds; once the time has passed, the link should no longer be valid for the expand command, but the original url could be minified again.

Command parameters:

- Parameter `--minify=https://www.example.com/path?q=search`
- Parameter `--expand=https://myurlshortener.com/fstp4`

If a complete URL has already been shortened and is not expired, the tool should return the same shortened URL. If a shortened URL doesn't exist or is expired, it should return an appropriate message.

Tech Stack:

- Do not use pre-existing libraries for generating shortened URLs.
- MongoDB for storage.
- Unit tests using pytest.
- Docker for containerization.
- Publish on GitHub

Exercise 2: Images CDN

Imagine you need to develop a CDN for storing and serving images via HTTP.

You are required to **design** a distributed system within an on-premise infrastructure, outlining **the scheme** that fulfills the following requirements:

- The cluster should consist of multiple nodes in permanent synchronization. In the event of uploading a new image (POST), it should be distributed to all cluster nodes as quickly as possible.
- When a GET request for an image is made, the queried node should be able to return the image even if it has not yet been synchronized on that specific node.
- In the event of a node going down, it should be able to resynchronize with the other cluster nodes as quickly as possible upon the next startup.

You can use any technology that you believe is most suitable for **designing** the required system.

Image deletion is not a consideration for the design of this system.